

7.2 Classwork: Log Challenge — Markscheme

1. [Maximum mark: 7]

Solve $\log_2(x - 3) = \log_4(x + 9)$.

Solution:

Change of base: $\log_4(x + 9) = \frac{1}{2} \log_2(x + 9)$ **M1**

$2 \log_2(x - 3) = \log_2(x + 9)$ **A1**

$\log_2((x - 3)^2) = \log_2(x + 9)$

Equating arguments: $(x - 3)^2 = x + 9$ **M1**

$x^2 - 6x + 9 = x + 9$

$x^2 - 7x = 0$ **A1**

$x(x - 7) = 0$

$x = 0$ or $x = 7$ **A1**

Domain requires $x - 3 > 0$, so $x > 3$. Reject $x = 0$. **R1**

$x = 7$ **A1**

2. [Maximum mark: 15]

(a) Series is geometric. [5]

(i) Show that $p = \pm \frac{1}{\sqrt{3}}$.

Solution:

$$\text{Common ratio: } \frac{p \ln x}{\ln x} = \frac{(1/3) \ln x}{p \ln x} \quad \text{M1}$$

$$p = \frac{1}{3p} \implies p^2 = \frac{1}{3} \quad \text{A1}$$

$$p = \pm \frac{1}{\sqrt{3}} \quad \text{AG}$$

(ii) Given $p > 0$ and $S_\infty = 3 + \sqrt{3}$, find x .

Solution:

$$r = \frac{1}{\sqrt{3}}, \quad a = \ln x$$

$$S_\infty = \frac{\ln x}{1 - \frac{1}{\sqrt{3}}} = \frac{\sqrt{3} \ln x}{\sqrt{3} - 1} = \frac{\sqrt{3}(\sqrt{3} + 1) \ln x}{2} = \frac{(3 + \sqrt{3}) \ln x}{2} \quad \text{M1}$$

$$\frac{(3 + \sqrt{3}) \ln x}{2} = 3 + \sqrt{3}$$

$$\ln x = 2 \quad \text{A1}$$

$$x = e^2 \quad \text{A1}$$

(b) Series is arithmetic with common difference d . [10]

(i) Show that $p = \frac{2}{3}$.

Solution:

$$p \ln x - \ln x = \frac{1}{3} \ln x - p \ln x \quad \text{M1}$$

$$(p - 1) \ln x = \left(\frac{1}{3} - p \right) \ln x$$

$$\text{Since } \ln x \neq 0: \quad 2p = \frac{4}{3} \quad \text{A1}$$

$$p = \frac{2}{3} \quad \text{AG}$$

(ii) Write down d in the form $k \ln x$.

Solution:

$$d = \left(\frac{2}{3} - 1\right) \ln x = -\frac{1}{3} \ln x \quad \mathbf{A1}$$

(iii) Sum of first n terms is $-3 \ln x$. Find n .

Solution:

$$S_n = \frac{n}{2} [2a + (n-1)d] \quad \mathbf{M1}$$

Substituting $a = \ln x$ and $d = -\frac{1}{3} \ln x$:

$$S_n = \frac{n}{2} \left[2 \ln x - \frac{n-1}{3} \ln x \right] \quad \mathbf{A1}$$

$$= \frac{n \ln x}{6} (7 - n) \quad \mathbf{A1}$$

Set equal to $-3 \ln x$ (and divide by $\ln x > 0$): **M1**

$$\frac{n(7-n)}{6} = -3$$

$$n^2 - 7n - 18 = 0 \quad \mathbf{A1}$$

$$(n-9)(n+2) = 0 \quad \mathbf{A1}$$

$$n = 9 \quad \mathbf{A1}$$

Reject $n = -2$ since $n \in \mathbb{Z}^+$.